

Grinding Machine: Working , Types & Applications

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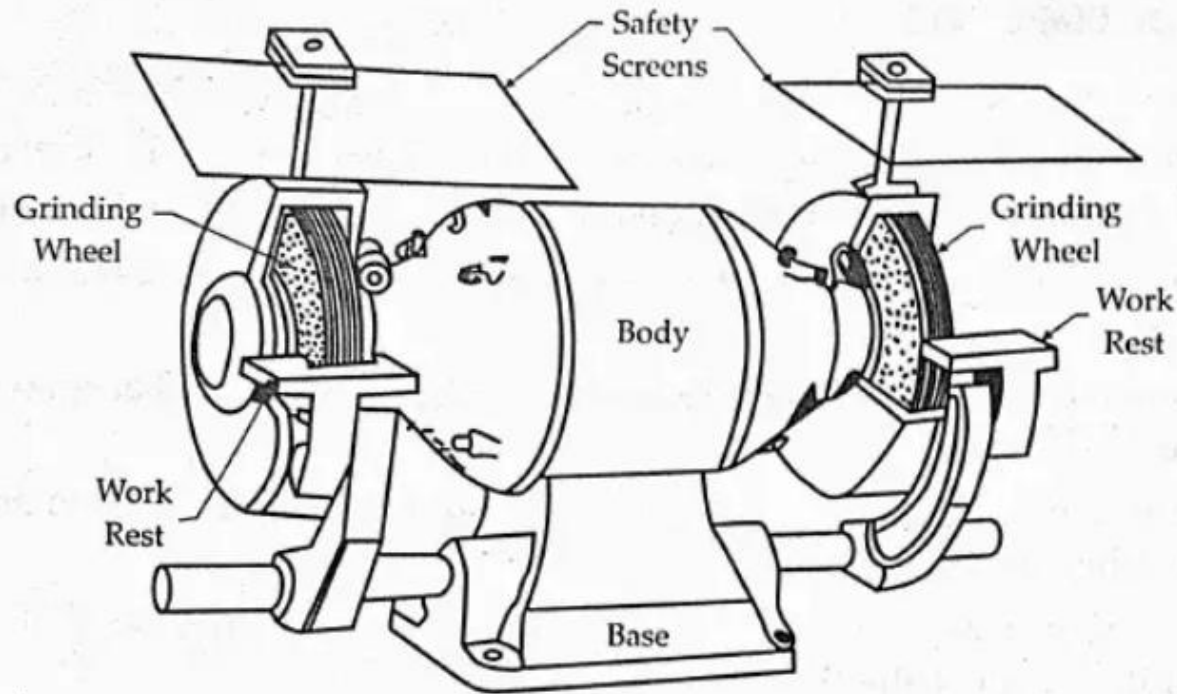
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01

Fundamentals of Grinding Machine



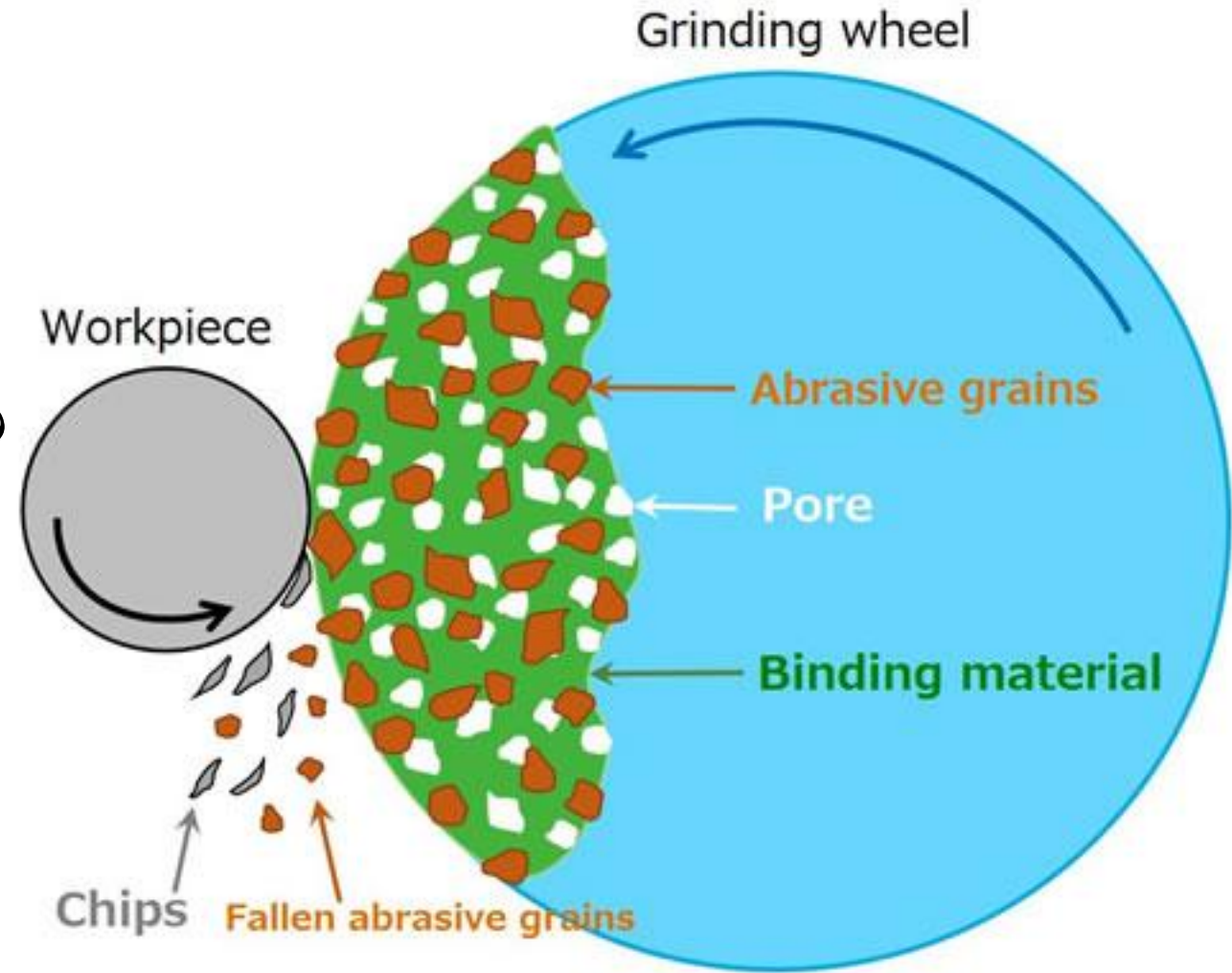
What is a Grinding Machine?

A Grinding machine is a power driven machine tool that uses a rotating abrasive wheel to remove material from the workpiece . It achieves extremely tight tolerances and fine surface finishes that conventional cutting tools cannot match.

Grinding is a critical finishing operation in manufacturing used after rough machining to achieve final dimensions and surface quality.

Working Principle

How Grinding
Removes Material?



Grinding removes material through a process called abrasive cutting.

Material Removal Process:

1. High-Speed Rotation

- The grinding wheel rotates at very high speed using an electric motor.

2. Abrasive Grains Act as Cutting Tools

- The wheel is made of many hard abrasive particles (like aluminum oxide or silicon carbide).
- Each grain works like a tiny sharp cutting edge.

3. Contact with Workpiece

- The rotating wheel comes into contact with the workpiece surface.
- Pressure is applied between the wheel and the material.

4. Chip Formation

- Abrasive grains penetrate the surface.
- Small chips of material are sheared off.
- Thousands of grains cut simultaneously.

5. Heat Removal

- Friction produces heat.
- Coolant is used to remove heat and protect the workpiece.

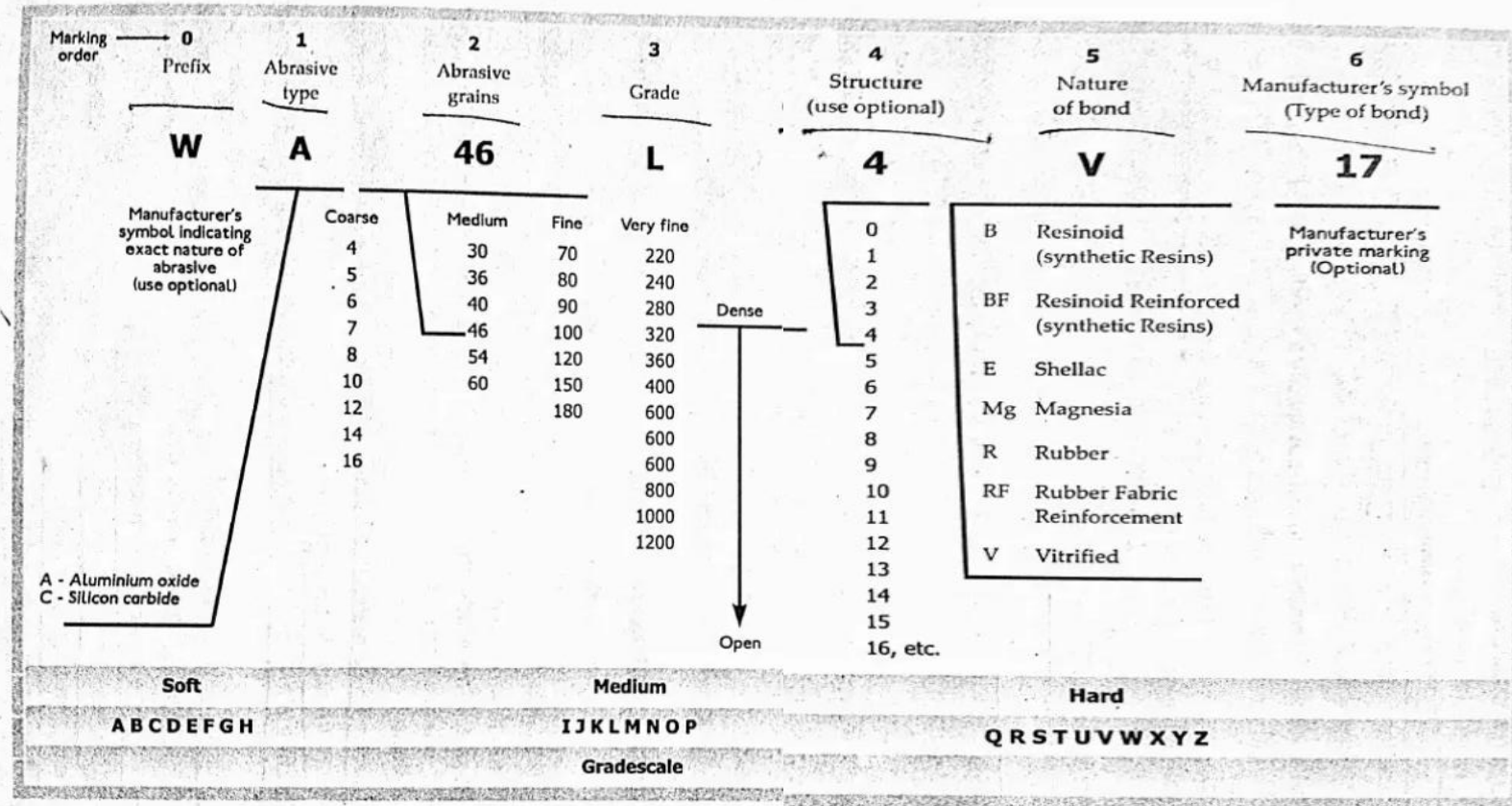


Fig. 12.6 Indian Standards system of marking Grinding Wheels (IS : 551-1989)

02

Constructions And Components

Main Parts of a Grinding Machine

1. Base

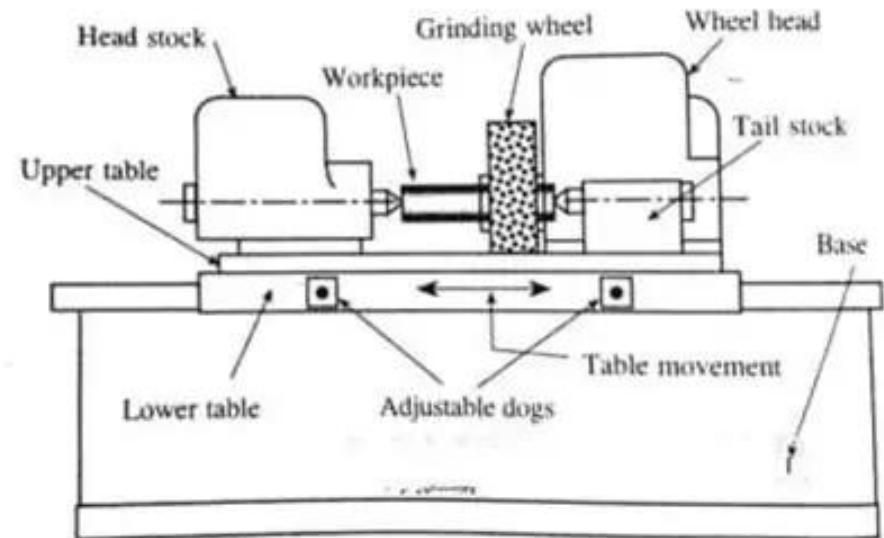
- The main supporting structure of the machine.
- Made of heavy cast iron to reduce vibration.
- Supports all other parts.

2. Grinding Wheel

- The most important part of the machine.
- Made of abrasive particles bonded together.
- Performs the cutting action and removes material.

3. Wheel Head

- Holds the grinding wheel and spindle.
- Provides rotational motion to the wheel.



Plain cylindrical grinding machine

Grinding Machine Schematic Diagram

4. Work Table

- Holds and supports the workpiece.
- Can move longitudinally or crosswise for feeding.

5. Work Holding Device

- Used to secure the workpiece.

Examples:

- Magnetic chuck
- Vice
- Centers or fixtures

6. Spindle

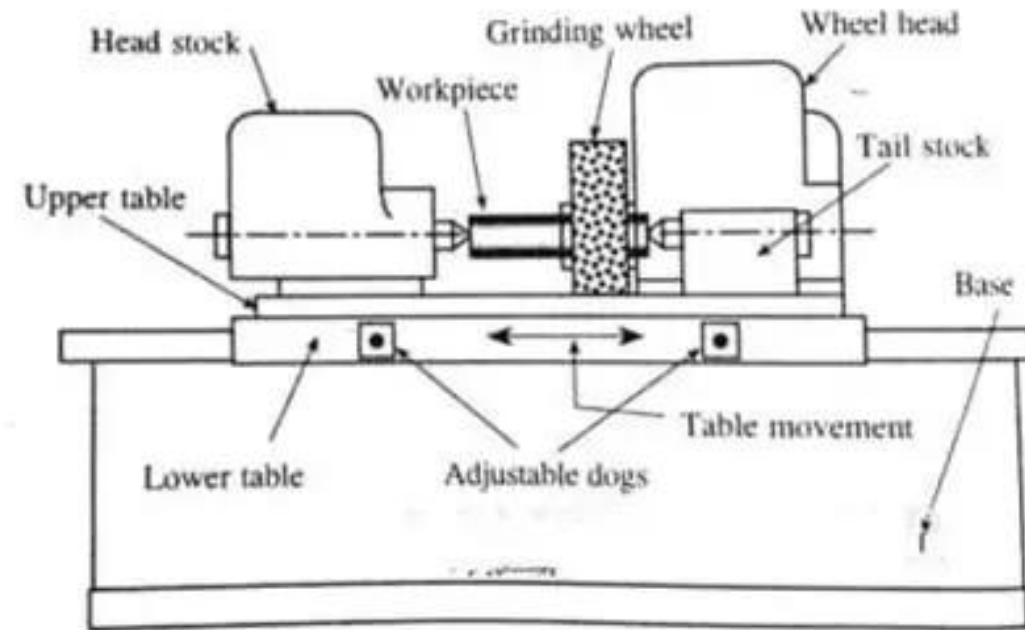
- Rotating shaft connected to the motor.
- Drives the grinding wheel at high speed.

7. Coolant System

- Supplies coolant during grinding.
- Reduces heat and improves surface finish.

8. Electric Motor

- Provides power for wheel rotation and table movement.



Plain cylindrical grinding machine

Grinding Machine Schematic Diagram

03

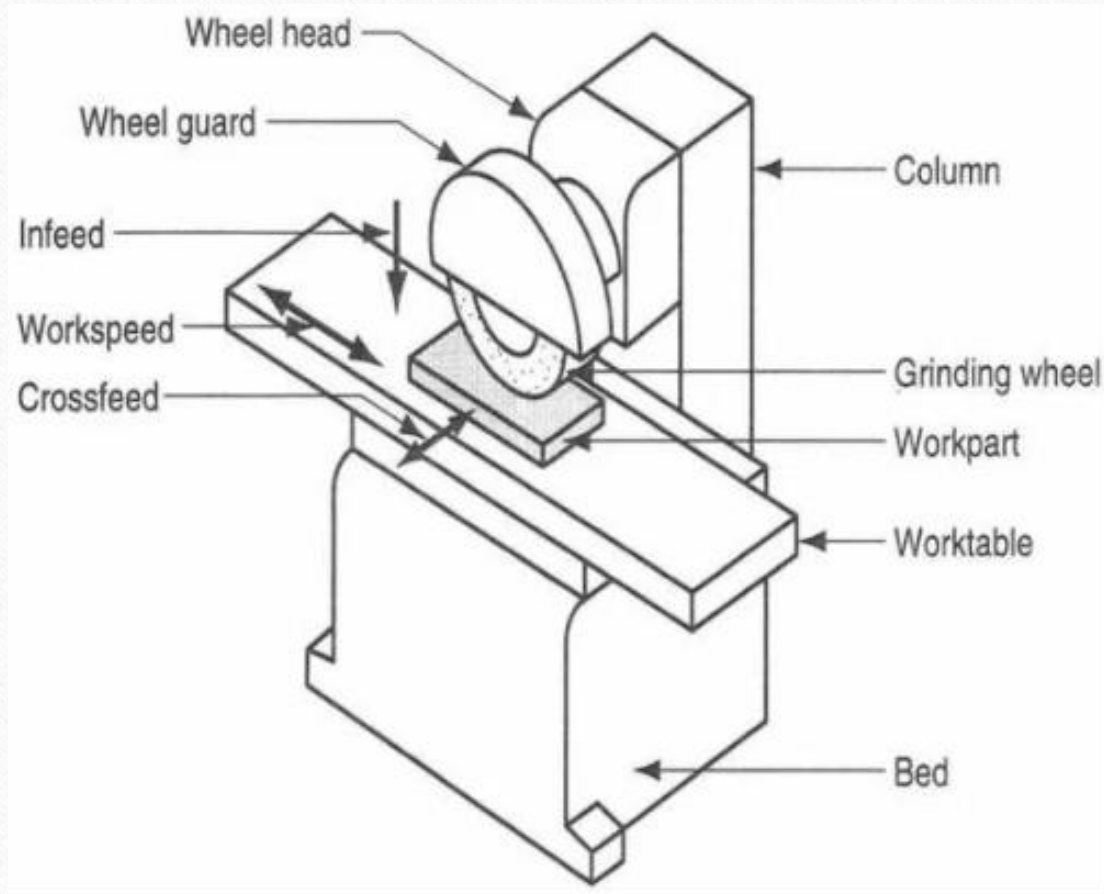
Types of Grinding Machine

Classification:

Types of Grinding Machine:

1. Surface Grinding Machine
2. Cylindrical Grinding Machine
3. Centreless Grinding Machine
4. Tool and Cutter Grinding Machine
5. Bench Grinding Machine
6. Portable Grinding Machine
7. Special Purpose Grinding Machines

Surface Grinding Machine

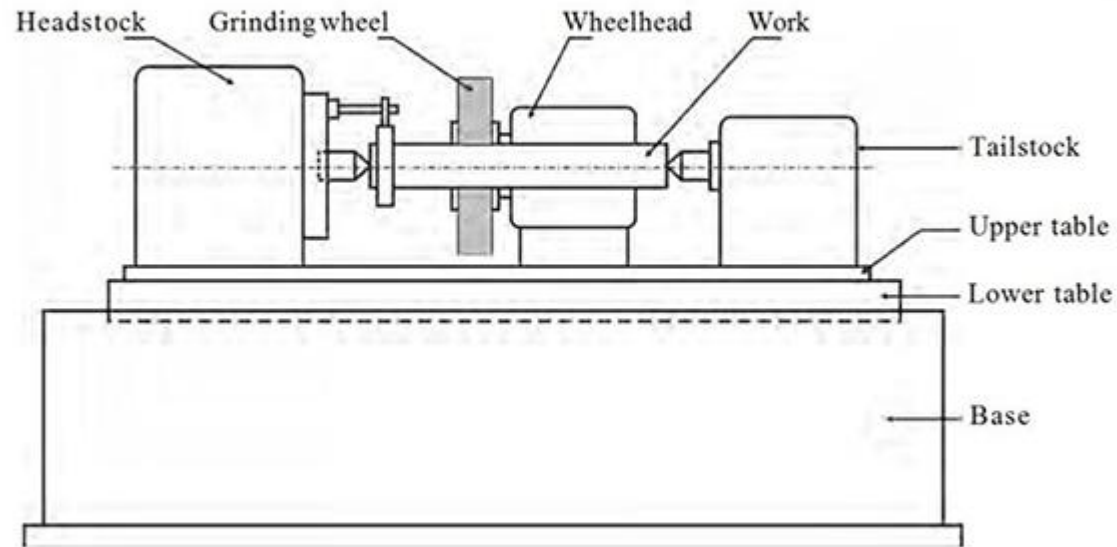


The workpiece is held on a reciprocating or rotatory table while a horizontal or vertical spindle-mounted wheel traverses across it producing a flat , smooth surface.

Applications:

- Die and Mold making
- Machine tool guideways
- Precision flat surfaces on gears and plates
- Aerospace structural components

Cylindrical Grinding Machine



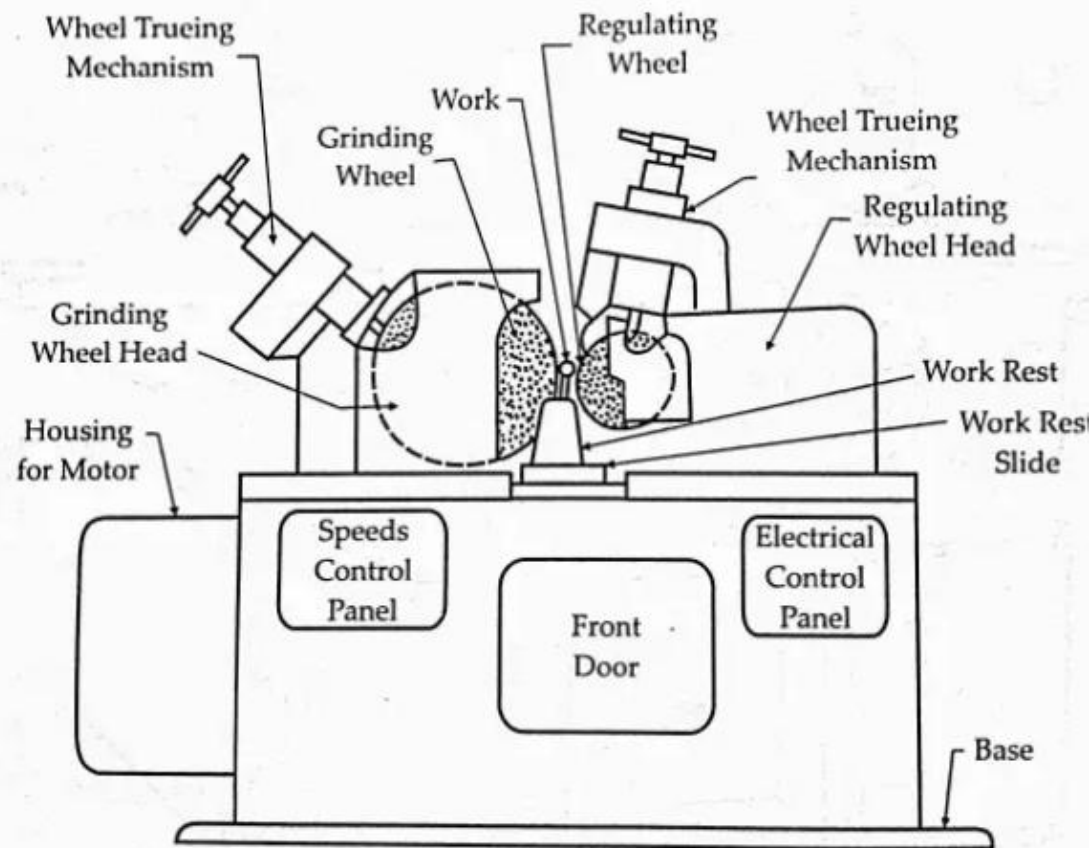
Cylindrical Grinding Machine

The workpiece rotates between centers or in a chuck while the grinding wheel traverses along its length . Can grind external diameters, internal bores , tapers, and shoulders with exceptional roundness.

Applications:

- Hydraulic Cylinder rod
- Automotive camshafts and crankshafts

Centerless Grinding Machine



The workpiece is not held between centers . It rests on a work rest blade, support between a large grinding wheel and a small regulating wheel . The regulating wheel controls feed rate and rotation.

Applications:

- No axial thrust on workpiece
- High production rate for long hours
- No setup time for centre's or chucks
- Excellent for through feed grinding

Tools And Cutters Grinders

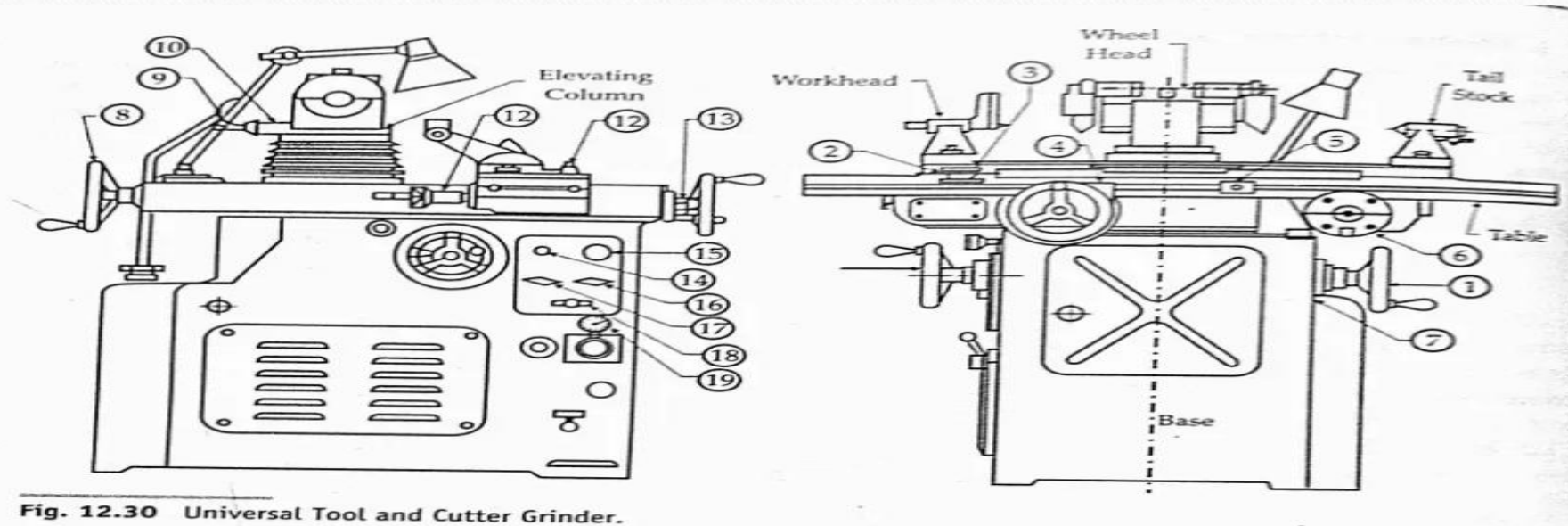
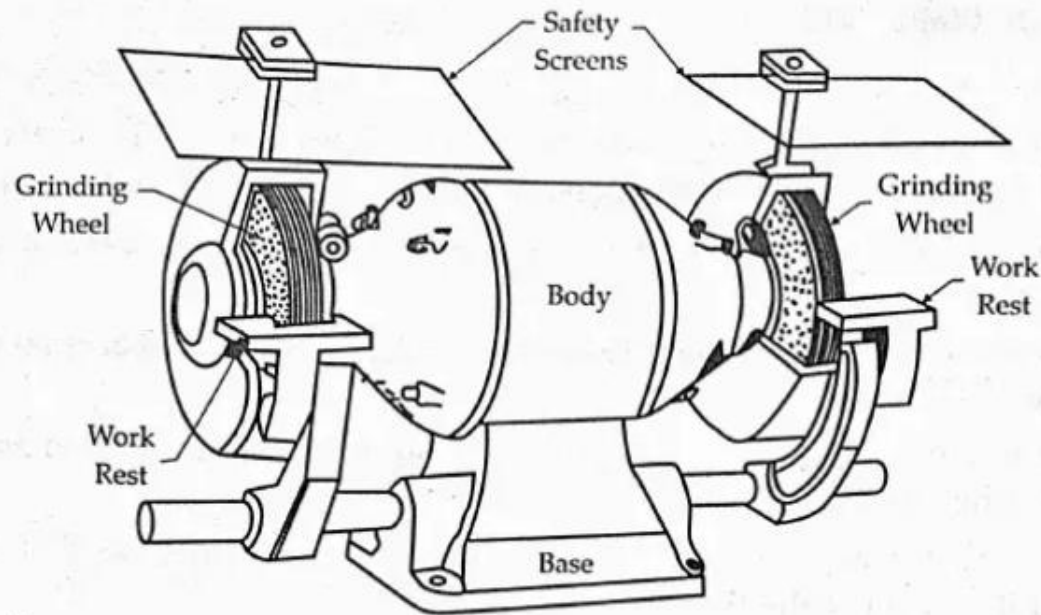


Fig. 12.30 Universal Tool and Cutter Grinder.

These machines are primarily intended for tool room work for grinding cylindrical and tapered multitooth cutting tools, like milling cutters, hobs, drills, reamers, taps, broaches, gear shaper cutters, etc.

They are also capable of doing light cylindrical, surface and internal grinding operations. They are made in various different designs. The most versatile and widely used form is, however, a Universal tool and cutter grinder, shown in Fig. 12.30.

Bench Grinding Machine



- Small grinding machine mounted on a bench.
- Used for rough grinding and tool sharpening.

Applications:

- Workshops and maintenance work.

Speed, Feed And Depth of cut

In case of grinding work, the term '**Cutting Speed**' refers to the *peripheral speed* of the grinding wheel relative to the speed of the workpiece. It is usually expressed in *metres per second* and can be expressed thus :

$$V = \frac{\pi DN}{1000 \times 60} \text{ m / sec}$$

where

V = Peripheral speed of the grinding wheel in m/sec.

D = Diameter of the grinding wheel in mm.

N = r.p.m. of the grinding wheel.

Feed

In cylindrical grinding, the term 'feed' denotes the axial movement of the workpiece in its each revolution. If ' f ' be the said feed, ' b ' the width of the face of the grinding wheel, then ;

$$f = C \cdot b \text{ mm/rev. of workpiece.}$$

where, C = a constant, depending upon the type of grinding being done,
i.e., rough grinding or finish grinding.

The values of ' C ' for the two types of grinding are as follows :

Rough grinding $\rightarrow$ 0.6 to 0.9

Finish grinding $\rightarrow$ 0.4 to 0.6

Now, the work or table travel can be calculated as follows :

$$\left. \begin{array}{l} \text{Work travel} \\ \text{or} \\ \text{Table travel} \end{array} \right\} = \frac{f \times n}{1000} \text{ metres/min}$$

where, f = feed in mm/rev.

n = r.p.m. of the workpiece.

Depth of Cut

It represents the thickness of the metal removed by the grinding wheel in one pass or one longitudinal traverse. It is given by :

$$t = \frac{D_1 - D_2}{2} \text{ mm}$$

where, $t = \text{depth of cut in mm}$

$D_1 = \text{workpiece dia. before grinding}$

$D_2 = \text{workpiece dia. after grinding.}$

The normal 'depth of cut' in grinding varies from 0.005 mm to 0.04 mm, depending upon the type of grinding, work material, bond, etc. Smaller values are adopted for finish grinding and higher values for rough grinding. The common ranges are :

For finish grinding = 0.005 mm to 0.015 mm

For rough grinding > 0.015 mm.

Machining Time in Grinding

12.35 MACHINING TIME IN GRINDING

The machining time (T) in cylindrical grinding can be calculated from the following relationship :

$$T = \frac{L \cdot P \cdot K}{f \cdot n} \text{ min}$$

where

L = Longitudinal travel in one pass (in mm)

P = Number of passes made

f = Longitudinal feed/rev. (in mm)

n = r.p.m. of the workpiece.

K = a constant (*Accuracy Factor*), depending upon the degree of accuracy and level of surface finish.

= 1.0 to 1.2 for rough grinding (commonly used value = 1.1)

= 1.3 to 1.7 for finish grinding (commonly used value = 1.4)

and

In case of *plumage cut* cylindrical grinding, the machining time (T) is given by :

$$T = \frac{a \cdot k}{f_c \cdot n} \text{ min}$$

where

a = grinding allowance on each side in mm.

f_c = cross feed in mm/rev.

n = r.p.m. of the workpiece

K has the same significance as in case of cylindrical grinding.

04

Performance Applications and safety

Industrial Applications of Grinding

Automotive

Crankshafts, camshafts, valve seats, and transmission gears

Aerospace

Turbine blades, landing gear components, and structural fittings

Tool Making

Sharpening drills, milling cutters, reamers, and broaches

General Manufacturing

Hydraulic rods, bearing races, spindles, and precision gauges

Advantages and Disadvantages:

Advantages:

1. High Accuracy
2. Excellent Surface Finish
3. Better Dimensional Control
4. Removes Very Small Amount of Material
5. Versatile Operation
6. Suitable for Mass Production
7. Produces Sharp Cutting Tools
8. Improves Product Quality

Disadvantages:

1. Heat Generation
2. Low Material Removal Rate
3. High Machine Cost
4. Wheel Wear
5. Skilled Operator Required
6. Risk of Overheating and Cracks

Essential Safety Precautions in Grinding Machine:

- 1. Wear Personal Protective Equipment (PPE)**
- 2. Inspect Grinding Wheel Before Use**
- 3. Proper Wheel Mounting**
- 4. Use Wheel Guard**
- 5. Maintain Proper Work Rest Distance**
- 6. Do Not Apply Excessive Pressure**
- 7. Stand Aside During Startup**
- 8. Use Coolant Properly**
- 9. Keep Workplace Clean**

Thank You